SHIV NADAR UNIVERSITY

Programme: **B.Tech** Discipline: **Computer Science and Engineering**
Exam**: Mid Semester** Year: **2016-2017**
Course Code: **CSD101** Course Title: **Introduction to Computing & Programming**
Date: **October 01, 2016** Time: **9:30 AM – 11:00 AM** Max. Marks: **50**

1. Write the output for the following format specification. **(2 Marks)**

(i) printf (" %7.2f ", 56.1624) (ii) printf (" %4d ", 35)

2. Determine the hierarchy of operations and evaluate the following expression: **(2 Marks)**

(i) kk = 3 / 2 * 4 + 3 / 8 + 3 (ii) float s = q*a/4–6/2+2/3*6/g (int q = 4, a=2, g=2)

3. **Fill in the blanks:** **(5 Marks)**

(i) The smallest data item in a computer, called a _____, can assume the value 0 or the value 1.
(ii) Flowchart symbols are connected by arrows called _____.
(iii) Small circle symbols in a flowchart are often called _____ symbols.
(iv) The _____ *operator* is the only ternary operator in C.
(v) Using an incorrect relational operator or using an incorrect final value of a loop counter in the condition of a while or for statement is a frequent cause of _____ errors.

4. **Tick the correct option:** **(5 Marks)**

(i) Which of the following is *not* a keyword?
a) int b) return c) if d) main

(ii) Which of the following is *true* of pseudocode programs?
(a) they are executed by the computer
(b) they help the programmer "think out" a program
(c) they typically include definitions and all types of statements
(d) all of the above are false

(iii)Various C statements enable you to specify that the next statement to be executed may be other than the next one in sequence. This is called _____.
a) change of order b) instruction skipping c) transfer of control d) rerouting

(iv)Which statement is true?
a) The expression ++(a + 1) adds 2 to a.
b) The expression **a multiplies a by 1.
c) The ANSI standard for the C programming language specifies the order in which each operator's operands will be evaluated.
d) The expression --(abc * 37) is a syntax error

(v) What is the final value of x after performing the following operations?

```
int x = 21;
double y = 6;
double z = 14;
y = x / z;
x = 5.5 * y;
```

5. Convert **$(10567)_8$** into its decimal equivalent. **(2 Marks)**
6. Draw the flowchart for the following program segment: **(5 Marks)**

```
int a, b, c, big;
a = 10, b = 20, c = 30;
big = (a > b ? (a > c ? a : c) : (b > c ? b : c));
printf("Largest number is %d", big);
```

7. A C program typically go through six phases during execution. Name them and briefly explain the function of each phase. **(6 Marks)**
8. Distinguish between the terms fatal error and nonfatal error. Why might you prefer to experience a fatal error rather than a nonfatal error? **(2 Marks)**
9. **True/False :** **(5 Marks)**

(a) When the printf function is called, it always begins printing at the beginning of a new line.
(b) The ***break*** statement is required in the ***default*** case of a ***switch*** selection structure.
(c) Conditions written inside decision symbols always contain arithmetic operators (i.e., +, -, *, /, and %).
(d) A machine language program requires translation before the program can be run on a computer.
(e) A valid C arithmetic expression containing no parentheses is evaluated from left to right.

10. What will be the output of the following code segments: **(6 Marks)**

(a)
```
void main()
{
int k = 35;
printf("%d \n %d \n %d", k = = 35, k = 50,
                                k > 40);
}
```

(b)
```
void main()
{
        int i = 4, j = -1, k = 0, w, x, y, z;
        w = i || j || k;
        x = i && j && k;
        y = i || j && k;
        z = i && j || k;
        printf("%d \n %d \n %d \n %d", w, x, y,
        z);
}
```

(c)
```
void main()
{
        int i = -3, j = 3;
        if(!i + !j * 1)
        printf("Massaro");
        else
        printf("Bennarivo");
}
```

(d)
```
void main()
{
 int k, num = 30;
 k = (num > 5 ? (num <= 10 ? 100 : 200) : 500);
 printf("%d", num);
}
```

(e)
```
void main()
{
        int j = 4;
        ( !j != 1 ? printf("Welcome") :
        printf("Good Bye"));
}
```

(f)
```
void main()
{
        int c = 3;
        switch(c)
        {
                Case '3': printf("Hi");
                        break;
                Case 3: printf("Hello");
                        break;
                default:
                        printf("How are you?");
        }
}
```

11. Rewrite the following code by changing the do-while loop into a for loop so that the functionality of the code will remain same. **(5 Marks)**

```
#include<stdio.h>
void main()
{
        int num;
        printf("Enter any number. Enter 999 to stop: \n");
        scanf("%d", &num);
        do
        {
                printf("%d \n", num);
                printf("Enter any number. Enter 999 to stop: \n");
                scanf("%d", &num);
        } while(num != 999);
}
```

12. Complete the following code so that it will produce the following pattern on output screen: **(5 Marks)**

```
#include<stdio.h>
void main()
{
        int i, j;
        for( i = ____ ; i < _____;  i++)
        {
                for( j = _____; j <= _____; j++)
                {
                        printf("%d", _____);
                }
                printf("\n");
        }
}
```

```
0
1 0
0 1 0
1 0 1 0
```